

Tanmay Mishra

Burlington, VT | tanmay.mishra@uvm.edu | [LinkedIn](#) | [Google Scholar](#)

Summary

Power systems and power electronics engineer with 10+ years of research and development experience in power systems dynamic and transient modeling, real-time simulation, hardware-in-the-loop and controller-in-the-loop testbed development, and inverter-based resource control. Proven track record of independent research, scientific publication in top IEEE and Elsevier journals, and presentation at national and international conferences. Experience with OPAL-RT real-time simulation, PSCAD and EMTP-RV user-defined models, PSS/E power system studies, and synchronous machine dynamics validated on hardware.

Education

Doctor of Philosophy — Electrical Engineering, Power Systems

2016 – October 2022

Indian Institute of Science (IISc), Bangalore, India | Thesis: Development of a Reconfigurable Synchronous Machine Emulation Platform

Master of Technology — Control Systems

2014 – 2016

Delhi Technological University, India | Ranked 3rd in batch | CGPA 8.48/10

Bachelor of Technology — Electrical and Electronics Engineering

2009 – 2013

National Institute of Technology (NIT) Hamirpur, India | Top 15% of batch

Skills

Power Systems Modeling and Simulation: PSCAD, EMTP-RV (user-defined inverter and plant controller models), PSS/E (load flow, N-1 contingency, IEEE 25-bus and IEEE 39-bus steady-state studies), MATLAB/Simulink, PLECS; transient stability, small-signal stability, fault analysis, voltage stability, weak-grid operation.

Real-Time Simulation and Hardware-in-the-Loop: OPAL-RT (RT-Lab, OP5033) — controller-in-the-loop and power-hardware-in-the-loop testbed development; inverter and site controller validation under grid disturbance and fault scenarios.

Power Systems Dynamic Modeling: Synchronous generator models (IEEE Std. 1.1, 2.1), IEEE Std. 421.5 excitation systems (AVR, limiters, PSS), turbine-governor models (steam single and double reheat, hydro), WECC-compliant composite load models (ZIP, induction motor), EMT modeling of centralized UPS data center based on PNNL reference manual.

Inverter-Based Resource Control: Grid-forming and grid-following inverters, virtual synchronous generator (VSG), PLL design, P-f and Q-V droop, Volt-Var (IEEE 1547), inner/outer current loops, virtual inertia — implemented in firmware and validated on hardware.

Embedded Firmware and DSP: TI C2000 — TMS320F28377D/S, F28335; Embedded C, Code Composer Studio; real-time inverter control, PWM, PLL, current and voltage loops.

Optimization and Grid Services: Python (Pyomo, NumPy, Pandas), Julia (JuMP, Gurobi, IPOPT); stochastic and MILP optimization for BESS dispatch, energy arbitrage, frequency regulation, capacity market bidding.

Programming: Python (primary), MATLAB scripting, Julia, Embedded C, C.

Experience

Research Scientist — Power Systems and Power Electronics

April 2024 – Present

University of Vermont (UVM), Burlington, VT

- Led development of kVA-scale IBR experimental testbed from inception to operating status: two 5 kVA inverters (GFM/GFL configurable), battery system with custom BMS, 15 kW electrolyzer load, bidirectional DC supply, regenerative grid simulator, and OPAL-RT OP5033 HIL platform — enabling CHIL validation of inverter and site controller performance under grid disturbance and fault scenarios.
- Developed grid-forming and grid-following inverter control algorithms from specification to hardware: inner/outer loop structure, PLL design, droop-based power control, virtual synchronous generator (VSG) and virtual inertia — evaluated across varying grid strength and weak-grid condition.
- Small-signal stability analysis of GFL inverters with PLL and Volt-Var control via state-space linearization: identified stability boundaries and dominant oscillatory modes across grid strength conditions.
- Developed stochastic optimization framework for multi-resource BESS dispatch in PJM frequency regulation markets: optimal capacity bidding, real-time power allocation, and flexibility quantification using Python (Pyomo) and Julia (JuMP/Gurobi/IPOPT).
- Built electro-thermal BESS and PV models for predictive dispatch: evaluated NLP and MILP scheduling algorithms across temperature extremes and battery C-ratings; quantified seasonal performance variability.
- Developing PSCAD-based models for centralized UPS-based data center architectures for EMT studies.
- Led \$4.5M DOE-funded Hybrid Energy Systems project as Senior Personnel: directed cross-disciplinary team, defined milestones, authored compliance deliverables for DOE stakeholder reviews.
- Co-mentoring PhD student on data-driven battery modeling, parameter estimation, and input-convex neural networks for BESS optimization.

Postdoctoral Researcher — Power Systems and Power Electronics

March 2023 – March 2024

University of Vermont (UVM), Burlington, VT

- Built Accelerated Test Laboratory (ATL) from scratch: hardware infrastructure for BESS characterization (charge/discharge cycling), inverter GFM/GFL validation, and real-time data acquisition — enabling dynamic model validation against hardware-in-the-loop behaviors.
- Designed and validated GFM/GFL inverter control algorithms; co-mentored PhD student on data-driven battery modeling and parameter estimation.

Graduate Research Assistant — Power Systems and Power Electronics

August 2016 – October 2022

Indian Institute of Science (IISc), Bangalore, India

- Developed reconfigurable synchronous generator emulation platform: dynamic inversion-based nonlinear controller emulating 100s-of-MVA generator dynamics on 5 kVA platform — validated on SMIB and WSCC 3-machine 9-bus systems under transient and fault disturbances.
- Designed and validated IEEE Std. 421.5 excitation systems on hardware: buck converter and H-bridge based excitation systems for micro-alternators; implemented AVR, PI control, excitation limiters, and time-constant regulation on TI TMS320F28377S/F28335 in Embedded C — tuned using Bode plot analysis.
- Developed and validated turbine-governor models (steam single/double reheat and hydro) on 11 kW DC motor drive: droop speed control, mechanical, electro-hydraulic, and hydraulic governors — closed-loop emulation of prime-mover dynamics and frequency response behavior.
- Developed and validated dynamic synchronous generator models (IEEE Std. 1.1, 2.1) in EMTP-RV: transient and small-signal stability analysis for SMIB and multi-machine systems; correlated EMT simulation against laboratory hardware measurements.
- Developed WECC-compliant composite load models (ZIP, induction motor) in EMTP-RV: validated experimentally via back-to-back converter emulator under varying grid conditions.
- Implemented IEEE-standard excitation systems and turbine-governor models in EMTP-RV: validated against experimental SMIB measurements; extended to 3-machine 9-bus network with inter-area dynamics validation.
- Developed back-to-back converter-based load emulator for WECC composite loads: SR-PLL synchronization, VSI inner current control, DC-link regulation implemented on TMS320F28377S in Embedded C.

Publications

Journal Papers

- **[J1] Tanmay Mishra**, Mads Almassalkhi, "Maximizing Power Flexibility of Hybrid Energy Systems for Capacity Market," IEEE Transactions on Smart Grid. (Submitted 2026)
- **[J2] Tanmay Mishra**, Dakota Hamilton, Mads Almassalkhi, "Optimal Bidding and Coordinated Dispatch of Hybrid Energy Systems in Regulation Markets," Electric Power Systems Research, Elsevier. (Submitted 2026)
- **[J3] Tanmay Mishra**, Durvesh Kalke, Gurunath Gurralla, Radhakant Padhi, "Synchronous Generator Emulation Using Output Matching Control Through Feedback Linearization," IEEE Transactions on Energy Conversion, 2025.
- **[J4] Tanmay Mishra**, Amritanshu Pandey, Mads Almassalkhi, "Predictive Optimization of Hybrid Energy Systems with Temperature Dependency," Electric Power Systems Research, Elsevier, 2024.
- **[J5] Tanmay Mishra**, Gurunath Gurralla, "Emulation of IEEE STD 421.5 / Industrial Excitation Systems Using a Micro-Alternator's Exciter," IEEE Transactions on Energy Conversion, 2023.

Conference Papers

- **[C1]** Arash Omid, **Tanmay Mishra**, Mads Almassalkhi, "Experimental Multi-Site Testbed for Advanced Control and Optimization of Hybrid Energy Systems," IEEE PES General Meeting, 2025.
- **[C2]** Arash Omid, **Tanmay Mishra**, Mads Almassalkhi, "Towards Input-Convex Neural Network Modeling for Battery Optimization in Power Systems," American Control Conference (ACC), 2025.
- **[C3] Tanmay Mishra**, Kiran Kumar Challa, Gurunath Gurralla, "Emulation of Steam Governor Turbine Models for Smart Grid Experimental Testbed," North American Power Symposium (NAPS), 2024.
- **[C4] Tanmay Mishra**, Gurunath Gurralla, "Excitation Limiters Design for a Micro-Alternator Buck Converter Based Static Excitation System," ICECET, 2022.
- **[C5] Tanmay Mishra**, Gurunath Gurralla, "Emulation of IEEE Excitation System Models Using Micro-Alternator's Excitation System," IEEE ISIE, 2020.
- **[C6] Tanmay Mishra**, Gurunath Gurralla, "Design and Implementation of IGBT-Based Static Excitation System with Buck Converter," IEEE PEDES, 2018.
- **[C7] Tanmay Mishra**, Ambrish Devanshu, Narendra Kumar, Ashish Kulkarni, "Comparative Analysis of Hysteresis Current Control and SVPWM on Fuzzy Logic Based Vector Controlled Induction Motor Drive," IEEE ICPEICES, 2016.

Awards and Honors

- GATE 2016 — All-India Rank 376 out of 150,000 candidates — 99.75th percentile
- MHRD PhD Fellowship — Ministry of Human Resource Development, Government of India (2016)
- RBCCPS Research Grant — Robert Bosch Centre for Cyber-Physical Systems, IISc Bangalore (2019)

Professional Service

- Peer Reviewer — IEEE Transactions on Power Systems, IEEE Transactions on Smart Grid, IEEE Transactions on Energy Conversion, Elsevier Electric Power Systems Research (EPSR), IEEE PESGM, NAPS, PowerCon, ISGT, PEDES
- Presenter — Power Systems Computation Conference (PSCC), IEEE PEDES, IEEE ISIE, IEEE NAPS.
- Founder Vice-Chair — IEEE Power and Energy Society Student Branch, IISc Bangalore (2022)